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Annual Review of Animal Biosciences

Domestic Animals as Potential Reservoirs of Zoonotic Viral Diseases

Oyewale Tomori[1] and Daniel O. Oluwayelu[2]

[1]African Centre of Excellence for Genomics of Infectious Diseases (ACEGID), Redeemer's University, Ede, Nigeria; email: oyewaletomori@gmail.com, oyewaletomori@run.edu.ng

[2]Department of Veterinary Microbiology and Centre for Control and Prevention of Zoonoses, University of Ibadan, Ibadan, Oyo State, Nigeria; email: ogloryus@yahoo.com, do.oluwayelu@ui.edu.ng

Annu. Rev. Anim. Biosci. 2023. 11:33–55

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-062922-060125

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Keywords

domestic animals, reservoir hosts, emerging viral zoonoses, endemic viral zoonoses, pandemics, One Health

Abstract

Zoonoses are diseases and infections naturally transmitted between humans and vertebrate animals. Over the years, zoonoses have become increasingly significant threats to global health. They form the dominant group of diseases among the emerging infectious diseases (EID) and currently account for 73% of EID. Approximately 25% of zoonoses originate in domestic animals. The etiological agents of zoonoses include different pathogens, with viruses accounting for approximately 30% of all zoonotic infections. Zoonotic diseases can be transmitted directly or indirectly, by contact, via aerosols, through a vector, or vertically in utero. Zoonotic diseases are found in every continent except Antarctica. Numerous factors associated with the pathogen, human activities, and the environment play significant roles in the transmission and emergence of zoonotic diseases. Effective response and control of zoonotic diseases call for multiple-sector involvement and collaboration according to the One Health concept.

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A transcendent moment nears upon the world for a microbial perfect storm. Unlike the meteorological perfect storm—happening just once in a century—the microbial perfect storm will be a recurrent event. The two events share a common feature; a combination of factors is the driving force behind each.

— *Microbial Threats to Health: Emergence, Detection, and Response* (Institute of Medicine, 2003)

1. INTRODUCTION

According to the World Health Organization (WHO) (1), a zoonosis is "any disease or infection that is naturally transmissible from vertebrate animals to humans." Such an infection is maintained in an animal population (a reservoir), which therefore serves as a continuous source of human infections. This encompasses infections humans acquire through direct contact with animals, as well as infections transmitted through indirect exposure routes, such as vector-borne or environmental and food system pathogens. Zoonotic viruses constitute an ongoing threat to human and animal health worldwide. Zoonotic pathogens, such as influenza and severe acute respiratory syndrome (SARS), account for most emerging infectious diseases (EID) in people (2), and more than three-quarters of emerging zoonosis pathogens originate from wildlife (3).

Zoonotic diseases have devastating impacts on human and animal health, livelihoods, and economies. In the past two to three decades, zoonotic diseases due to viral pathogens have taken over the disease landscape worldwide, causing varying degrees of morbidity and mortality, as well as economic and productivity losses. For instance, the ongoing coronavirus disease 2019 (COVID-19) pandemic caused by SARS coronavirus-2 (SARS-CoV-2) has been classified as a zoonotic disease prematurely, because no animal reservoir has yet been found. It is proposed that COVID-19 should be classified an "emerging infectious disease (EID) of probable animal origin" (4). Over the 30 months since COVID-19 was first reported in Wuhan, China, there have been more than 543 million cases and more than 6.3 million deaths globally (5). Other examples of zoonoses are the extremely rare sylvatic spillover cases of Marburg, Crimean-Congo hemorrhagic fever (CCHF), and henipaviruses from wild animals and Rift Valley fever (RVF) and Middle East respiratory syndrome (MERS) from livestock. Lassa fever is acquired from a peri-domestic rodent, whereas the sources of Ebola virus, SARS, and SARS-CoV-2 remain enigmatic, with most human infections from these diseases being acquired through human–human transmission (4).

The global human population is expected to increase from 6.5 billion in 2008 to 9.2 billion by 2050 (6). This poses enormous challenges for food production in general, with an increased demand for up to 50% more food expected by 2030, especially for meat and dairy products (7, 8). To meet this demand, two methods of farm production are being adopted, which bring with them their own risks in terms of the introduction and transmission of infectious diseases.

One is very-low-intensity subsistence livestock farming, particularly of poultry, sheep, and goats, which operates in the rural households of the poor parts of the world. These animals are often kept under scavenging conditions, with little attention to disease control, housing, or feed supplementation (9). The animals, which suffer a high burden of endemic diseases, are in close contact with other livestock species and humans and potentially in contact with a variety of non-domestic animals. These conditions favor closer human–animal contact and increased possibility for zoonoses transmission. At the other end is the highly organized and intensive sector of the poultry industry, in which large numbers of birds are kept in a single shed. This intense husbandry practice requires control of infectious diseases via the administration of vaccines and drugs. These settings favor the development of antimicrobial resistance. Regarding zoonoses, this system facilitates the emergence of new pathogens, or a new variant of an old pathogen with the potential to

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spread rapidly and devastate national flocks, or the transmission of the strains of highly pathogenic avian influenza (AI) (10, 11).

2. VIRUSES AS PATHOGENS OF ZOONOTIC DISEASES

Zoonotic diseases caused by viral, bacterial, fungal, protozoan, and parasitic agents shared in nature by man and animals (domestic and wildlife) constitute major public health concerns worldwide. The incidence of EID events has increased since 1940, reaching a maximum in the 1980s. From decade to decade, the proportion and severity of zoonotic diseases affecting man have been increasing. Approximately 60% of infectious agents are known to be pathogenic for humans, 75% of pathogens associated with emerging diseases are zoonotic, and the risk of zoonoses is predicted to continue to increase. Only approximately 25% of these pathogens have their origin in domestic animal species, whereas the rest originate from wildlife species (2, 3, 12). RNA viruses pose particularly high zoonotic risks because they can emerge and spread rapidly, and a statistical analysis of 146 livestock viruses indicated that the ability of a virus to replicate in the cytoplasm (without nuclear entry) is the strongest single predictor of cross-species transmission and the ability to infect humans (13).

Viral zoonoses constitute a major threat to animal and public health worldwide, and their effective control is crucial for agronomic health, for safeguarding and securing national and international food supplies, and for alleviating rural poverty in developing countries. Some devastating livestock diseases are endemic in many parts of the world, and threats from old and new pathogens continue to emerge, with changes to global climate, agricultural practices, and demography presenting conditions that are especially favorable for the spread of arthropod-borne diseases into new geographical areas. Zoonotic infections that are transmissible either directly or indirectly between animals and humans are on the rise and pose significant additional threats to human health (14). **Table 1** provides information on geographic distribution, animal hosts/reservoirs, and means of transmission of some major viral zoonoses.

3. ANIMALS AS RESERVOIRS OF VIRAL ZOONOSES

According to Haider et al. (4), zoonotic viral richness correlates strongly with mammalian species diversity and abundance, with domestic species harboring, on average, 19.3 zoonotic viruses compared to an average of 0.23 for free-ranging wildlife. Over the years, the term natural reservoir has been described in several ways, with no specific agreed-upon definition, due majorly to the large numbers of infectious microorganisms capable of causing diseases. The natural reservoir refers to the population of organisms or the specific environment in which an infectious pathogen naturally lives and reproduces, or upon which the pathogen primarily depends for its survival, and from where it is transmitted to the target population. It is usually a living host of a certain species (human or animal), inside of which a pathogen survives, often without causing disease for the reservoir itself, or it may be an environment external to an organism, such as contaminated air or water (14, 15). Natural reservoirs of infectious disease pathogens are the habitats in which the agent normally lives, grows, and multiplies. They include humans and animals (animate) and the environment (inanimate). The reservoir may or may not be the source from which the agent is transmitted to a susceptible host (15). Reservoirs may include one or more different species; may target the same or a different species; and may include vector species (14), which are distinct from natural reservoirs. Importantly, species considered reservoirs for a given pathogen may not experience symptoms of disease when infected by the pathogen.

Human reservoirs refer to persons infected by specific pathogens that live on or within the human body. For example, humans are known reservoirs for measles, mumps, various respiratory

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Table 1 Major viral zoonoses: geographical spread, animal hosts/reservoirs, and transmission routes

Geographical spread	Virus	Animal hosts/reservoirs	Means of transmission
Global	Rabies	All warm-blooded animals, especially dogs	Bite of infected animal
	Hantavirus	Rodents	Aerosolized rodent urine/feces; saliva or bite of infected rodent
	Yellow fever virus	Nonhuman primates	Bite of infected mosquitoes (primarily *Aedes* spp.)
	Chikungunya virus	Nonhuman primates, rodents, birds, small mammals	Bite of infected mosquitoes (primarily *Aedes* spp.)
	Crimean-Congo hemorrhagic fever virus	Domestic ruminants (cattle, sheep, goats)	Bite of infected ticks (mainly *Hyalomma* spp.); direct contact with blood or tissues of infected ticks, viremic livestock or viremic humans
	West Nile virus	Birds	Bite of infected mosquitoes (primarily *Culex* spp.)
Africa/Middle East	Rift Valley fever virus	Domestic ruminants (cattle, sheep, goats, camels)	Bite of infected mosquitoes (primarily *Aedes* spp.); contact with blood or organs of infected animals; inhalation of aerosols during slaughter of infected animals; consumption of unpasteurized milk
	Ebola virus	Fruit bats, nonhuman primates, antelopes, porcupines	Contact with blood, secretions, organs, or other bodily fluids of infected animals or humans
	Marburg virus	Fruit bats (*Rousettus aegyptiacus*)	Contact with infected bat feces or aerosols or with blood, secretions, organs, or other bodily fluids of infected people, and with surfaces and materials contaminated with these fluids
	Lassa fever virus	*Mastomys natalensis* rodents	Rodent urine or feces; nosocomial spread
	Monkeypox virus	Rodents, nonhuman primates	Contact with blood, bodily fluids, or lesions of infected animals or with respiratory secretions, skin lesions of infected persons, or contaminated objects
	Middle East respiratory syndrome virus	Dromedary camels	Direct or indirect contact with infected dromedary camels; close contact with infected persons; contact with fomites
Americas	Equine encephalitis viruses: EEEV, WEEV, VEEV	Wild birds; equids (for VEEV only)	Bite of infected mosquitoes (*Culex*, *Culiseta*, and *Aedes* spp.)
	Colorado tick fever virus	Wild rodents	Bite of infected Rocky Mountain wood ticks (*Dermacentor andersoni*)
	Zika virus	Nonhuman primates	Bite of infected mosquitoes (primarily *Aedes* spp.); transplacental; sexual intercourse; transfusion of blood and blood products; organ transplantation

(*Continued*)

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Table 1 (*Continued*)

Geographical spread	Virus	Animal hosts/reservoirs	Means of transmission
Europe	Tahyna virus	Small mammals (hares, rabbits, hedgehogs, rodents); large mammals (red deer, roe deer, wild boar)	Bite of infected *Aedes vexans* mosquitoes
	Tick-borne encephalitis virus	Small rodents (voles, mice); wild and domestic mammals (foxes, bats, hares, deer, wild boar, sheep, cattle, goats)	Bite of infected *Ixodes* and *Dermacentor* spp. ticks; consumption of unpasteurized milk
	Usutu virus	Wild and domestic birds (blackbirds, doves, jays, magpies, partridges, owls, sparrows)	Bite of infected mosquitoes (mainly *Culex* spp.)
Asia	Influenza viruses	Wild waterfowl, domestic poultry, swine, horses, dogs, bats	Inhalation of droplets from infected persons; direct contact with infected animals or environments contaminated with their nasal secretions, saliva, or feces
	Severe acute respiratory syndrome coronaviruses	Variety of domestic and wild animals (palm civets, bats? pangolins? dogs? cats? etc.)	Contact with droplets from infected persons; contact with contaminated surfaces/fomites
	Japanese encephalitis virus	Pigs, wading birds	Bite of infected *Culex* spp. mosquitoes
	Nipah virus	Fruit bats, pigs	Direct contact with infected animals or their body fluids; consumption of foods contaminated by body fluids of infected animals; contact with infected persons or their body fluids
Australia	Hendra virus	Fruit bats, horses	Exposure to body fluids and tissues/excretions of infected horses
	Kunjin virus	Wading birds	Bite of infected mosquitoes (primarily *Culex annulirostris*)

Abbreviations: EEEV, Eastern equine encephalitis virus; VEEV, Venezuelan equine encephalitis virus; WEEV, Western equine encephalitis virus.

pathogens, sexually transmitted diseases, and smallpox virus (15). Animals serve as reservoirs for many pathogens that occasionally jump species and infect humans. Animal reservoirs consist of domesticated and wild animals infected by pathogens (14, 15), and numerous zoonotic diseases have been traced back to these animals. For example, in nature, lyssaviruses (including rabies) are associated with specific mammalian reservoir hosts, bats and carnivores (16). The natural reservoirs of Ebola and Marburg viruses remain a mystery, although both viruses are suspected to be maintained in exposed small animals, with bats heading the list of suspects (17–19). Animals are known to play significant roles in the transmission of zoonotic viral diseases to humans, especially occupationally exposed persons including veterinarians, butchers and slaughterhouse workers, livestock farmers, researchers, pet owners (through bites or scratches of owners of indoor pets), zookeepers, and hunters.

Although this review is focused on the role of domestic animals as reservoir hosts of zoonotic viral diseases, transmission can also occur through vectors such as rodents and arthropods—ticks, mosquitoes, and *Culicoides* midges. Whereas ticks and mosquitoes are widely reported as vectors

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of major viral zoonoses, little is known about the role of *Culicoides* in the transmission of viruses of public health importance (20). Although more than 50 viruses have been isolated from *Culicoides* spp. worldwide, most of them are of veterinary importance and belong to the families *Reoviridae*, *Rhabdoviridae* and *Peribunyaviridae* (21, 22). For example, Shuni virus, a member of the family *Peribunyaviridae*, which causes congenital malformations in cattle and sheep as well as neurological symptoms in cattle and horses (23, 24), was first isolated from healthy cattle in Nigeria (25) as well as from *Culicoides* biting midges and mosquitoes in Africa (26). However, it is suspected to possess zoonotic potential, as it was isolated from a febrile child in Nigeria (27), and specific antibodies were found in large animal veterinarians in South Africa (28).

In addition, several bat species have been reported as reservoirs of zoonotic viruses, such as rabies and other lyssaviruses, Hendra virus (HeV) and Nipah virus (NiV), Menangle virus, Tioman virus, and SARS-CoV-like viruses (29, 30). Several reasons have been adduced as explanations for bats harboring many viruses. First, the large number of bat species and individuals contribute to the existence of many bat-borne diseases, and second, some physiological peculiarities of bats make them good reservoir hosts (31). These peculiarities include their food choices, population structure, ability to fly, seasonal migration and daily movement patterns, torpor and hibernation, long life span, and roosting behaviors (29). Other animals that harbor and transmit zoonotic pathogens include the multimammate rat, *Mastomys natalensis*, which serves as reservoir for Lassa fever virus, excreting large doses of the virus through urine and feces. Humans become infected directly through the fecal–oral route or via inhalation of contaminated air (32). The deer mouse that transmits Sin Nombre virus, the agent of hantavirus pulmonary syndrome (33), and known monkey reservoirs of herpes B virus (34) are other pertinent examples.

Environmental reservoirs such as plants, soil, and water can also serve as reservoirs for some infectious agents. Many fungal agents, such as those that cause histoplasmosis, live and multiply in the soil. Outbreaks of Legionnaires' disease are often traced to water supplies in cooling towers and evaporative condensers, reservoirs for the causative organism *Legionella pneumophila* (15).

Against the background of zoonoses accounting for approximately 60% of new disease introductions (2, 3, 8), the world's megacities provide an obvious focus of attention, as they typically constitute melting-pot environments for the mixing of human and animal infectious diseases and their potential rapid spread, both locally and internationally (9).

In 2000, 18 megacities had populations of more than 10 million inhabitants; by 2025, Asia alone is expected to have at least 30 megacities; and by 2030, more than 2 billion people in the world will be living in slums associated with cities (35). The emergence of the SARS and, recently, SARS-CoV-2 viruses and their rapid international spread provide examples of the transmission of new diseases from megacities (9, 36).

Identification of the natural reservoirs of infectious pathogens has been used in the control of large outbreaks of disease in humans and domestic animals, especially those diseases for which no vaccine exists. In principle, zoonotic diseases can be controlled by isolating or destroying the pathogen's reservoirs of infection. For instance, the mass culling of animals confirmed or suspected as reservoirs for human pathogens, such as birds that harbor AI, has been effective at containing epidemics in many parts of the world (37, 38).

4. EMERGING AND ENDEMIC VIRAL ZOONOSES AND DOMESTIC ANIMAL RESERVOIRS

The WHO's Global Burden of Disease (GBD) project measures burden of disease using the disability-adjusted life years (DALY) metric. The DALY is a time-based measure that adds together years of life lost owing to premature mortality with the equivalent number of years of life

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lived with disability or illness (39). Emerging and endemic viral zoonoses, including AI, Ebola virus disease, Lassa fever, rabies, swine influenza, CCHF, RVF, and, lately, COVID-19, have become major contributors to the overall incidences of morbidities, mortalities, massive economic losses in the livestock industry, and regional and global health emergencies. Unlike that of emerging viral zoonoses that occur in the developed world, where they pose a direct threat to the well-being of the rich, the burden of endemic zoonoses falls particularly heavily on the poor, as they are often compelled to live in close contact with their animals and so are more likely to become infected from these disease reservoirs (40).

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4.1. Emerging Viral Zoonoses

Many emerging zoonoses are caused by viruses that are perceived as the greatest threat to global health security because of their diversity and evolutionary properties. The last two decades have witnessed the emergence of several viral zoonoses that constitute public health threats across the globe, with most of them having animals as their reservoirs. According to MacLachlan & Dubovi (41), an emerging viral disease is one that is newly recognized or newly evolved, or that has occurred previously but shows a rapid increase in incidence or expansion in geographical, host, or vector range. Between 1985 and 2004, a new virus emerged virtually every year (40). Indeed, the newly emerging diseases that resulted in major epidemics in that period were caused mostly by zoonotic viruses that had their origins in animals, including SARS from horseshoe bats, NiV from *Eidolon helvum* fruit bats, and H5N1 AI from poultry (40). An estimated 60–80% of newly emerging infections are zoonotic in origin and thus depend on animal reservoirs for persistence, especially at the initial stages (3). Apart from wildlife, domestic animals also constitute important reservoirs of emerging viral zoonoses, with the anthropogenic influence on ecosystems determining the level of risk that operates at the human–animal–environment interface in zoonotic disease emergence. Several factors can perpetuate the spread and persistence of newly emerged (or reemerged) pathogens, including climate change, expanding global human population and urbanization, international trade and travels, proliferation of reservoir populations, and antimicrobial drug use, as well as anthropogenic activities such as intensive livestock keeping and land-use change occasioned by deforestation, mining and agriculture, and urban and industrial development (42–44). A higher diversity of zoonotic agents is found where the diversity of host animals is higher (45), and higher animal host and pathogen species diversity is associated with an increased risk of disease emergence (46).

4.1.1. Avian influenza (H5N1). AI is a poultry disease caused by type A influenza virus, a member of the genus *Alphainfluenzavirus* in the family *Orthomyxoviridae* (47). It is one of the greatest public health concerns that has emerged from animal reservoirs in recent years. AI viruses (AIVs) are believed to be in evolutionary stasis in their natural wild bird hosts, in which the virus and the host tolerate each other (48). However, the current outbreaks detected in poultry in many Asian, European, and African countries are of concern both to the poultry industry, in which they produce an economically devastating disease, and to public health (49). Presently, based on the antigenic characteristics of the immunodominant viral hemagglutinin and neuraminidase proteins, 18 hemagglutinin and 11 neuraminidase subtypes of influenza A viruses are recognized (50, 51). In 1997, AIV A/H5N1 subtype emerged in Hong Kong, causing high (75–100%) mortality in commercial chicken flocks and 6 fatalities out of 18 human cases recorded (52). The virus has become endemic in poultry in Asia and has now spread to Africa and Europe (53, 54). Epidemiologic and molecular evidence suggest that poultry was the source of the 1997 H5N1 outbreak in humans (55).

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4.1.2. Zoonotic coronaviruses: severe acute respiratory syndrome coronaviruses and Middle East respiratory syndrome coronavirus. Although coronaviruses cause respiratory and intestinal infections in animals and humans (56), they were not considered to be highly pathogenic to humans until the SARS outbreak in 2002–2003 in Guangdong Province, China (57). The *Coronaviridae* family is composed of four genera: alpha- and betacoronaviruses, which are found in mammals, and gamma- and deltacoronaviruses, which are found in birds (58). Six recognized coronavirus types can infect humans, of which three betacoronaviruses, namely, SARS-CoV, MERS coronavirus (MERS-CoV), and SARS-CoV-2, are zoonotic and have caused highly pathogenic respiratory infections globally (59, 60). SARS-CoV was the first severe and readily transmissible new disease to emerge in the twenty-first century and showed a clear capacity to spread along the routes of international air travel (61). A decade after SARS-CoV, another highly pathogenic coronavirus, MERS-CoV, emerged in the Middle East (62); unlike SARS-CoV, its transmission is geographically limited, and reported cases have often stemmed from outbreaks within Middle Eastern countries or recent travel to the region (63). Earlier studies suggested that SARS-CoV and MERS-CoV originated from bats (64, 65) but were transmitted to humans through intermediate amplification hosts, i.e., masked palm civets and dromedary camels, respectively (64, 66). Additionally, the pandemic caused by the recently emerged SARS-CoV-2 presumably originated from a bat coronavirus with 96% nucleotide similarity to the SARS-CoV-2 detected in a bat from Yunnan Province (SARS-CoV; RaTG13) (67). Also, some reports suggest that the Malayan pangolin (*Manis javanica*) may serve as an intermediate host of SARS-CoV-2 (67, 68).

All human coronaviruses (HCoVs) reportedly have animal origins based on current sequence databases. For example, SARS-CoV, MERS-CoV, and HCoV-NL63 and HCoV-229E are considered to have originated in bats, whereas HCoV-OC43 and HCoV-HKU1 possibly originated from rodents (69). Also, viruses belonging to lineage D of *Betacoronavirus*, considered to be restricted to fruit bats (70), were found recently in fecal samples collected from tree foliage in a roost of *E. helvum* fruit bats in southwestern Nigeria (71). Although coronaviruses circulate widely in wildlife, the bat-to-human transmission and the role of the intermediate host in transmission are still unclear (72). Cui et al. (73) noted that domestic animals may play important roles as intermediate hosts that aid virus transmission from natural hosts to humans. However, more research is required to elucidate the role of domestic livestock as potential reservoirs capable of transmitting coronaviruses to humans.

4.1.3. Nipah and Hendra viruses. *Henipavirus* is a genus in the family *Paramyxoviridae*, which includes two highly pathogenic zoonotic viruses: NiV and HeV. Both viruses cause severe encephalitis in humans, with case fatality rates ranging from 40% to 100% (74, 75). NiV was first diagnosed in Malaysia in 1999 and has caused serious disease in humans and livestock (76, 77). Transmission to humans occurs primarily through contact with infected pigs (76), although transmission via ingestion of food products contaminated with the saliva or urine of pteropid fruit bats has also been reported (78). Serologic studies suggested that, apart from bats, natural infections with NiV occurred in dogs, cats, and horses in the outbreak area in Malaysia (79). Other domestic animals, including cattle, were associated with illness caused by NiV in the outbreaks in Bangladesh (80).

In 1994, HeV, described initially as an equine morbillivirus, emerged in two outbreaks of severe respiratory disease in thoroughbred horses in Hendra, a suburb of Brisbane, and in Queensland, Australia, with human fatalities resulting from exposure to infected horses (81, 82). Experimentally, HeV has been reported to infect cats and horses (83), and Li et al. (84) showed that pigs are susceptible to HeV infections and could potentially play a role as intermediate hosts in transmission to humans. HeV infections have also been reported in dogs (85). In addition, non-neutralizing

antibodies against henipaviruses have been detected in pigs (86). Further research is needed to fully elucidate the role of domestic animals in transmission of henipaviruses.

An interesting but dangerous dimension to the topic of domestic animal reservoirs and viral zoonoses is the association of these diseases with pet ownership. According to Chomel & Sun (87), due to increasing urbanization, pets have become an essential part of families in many developed countries, sharing human lifestyles, bedrooms, and beds, with an estimated 14–62% of pet owners permitting dogs and cats on their beds. Also, based on the 2021 People's Dispensary for Sick Animals (88) Animal Wellbeing Report, 9.6 million pet dogs are owned by 26% of the UK adult population, 10.7 million pet cats are owned by 24% of the UK adult population, and approximately 900,000 pet rabbits are owned by 2% of UK adults. Additionally, in the Netherlands, with a pet population of ≈2 million dogs and 3 million cats, 50% of pet owners interviewed allowed the pet to lick their face; 60% of pets visited the bedroom; 45% of dogs and 62% of cats were allowed on the bed; and 18% and 30% of dogs and cats, respectively, were allowed to sleep with the owner in bed (89). These various types of pet contact have been reported as routes of transmission of different viral, bacterial, and parasitic zoonoses, including rabies, plague, staphylococcosis, toxocariasis, and cryptococcosis (87). In developing countries, levels of awareness and keeping of domestic animals as pets have been increasing. For instance, in Nigeria, there is an upsurge in keeping of exotic dogs as pets, and this has led to the establishment of close bonds between these dogs and their owners (90). In addition, several households, organizations, and individuals raise these dogs as guards and for breeding (economic) purposes. Similarly, Nigerian village dogs, which are used mostly for hunting purposes, enjoy a close relationship with the hunters. However, public health risks, including transmission of zoonoses such as rabies, may be associated with such practices. For example, the findings of a survey of rabies virus antibodies in confined, hunting, and roaming dogs in southwestern Nigeria revealed the presence of antibodies in only 13% of a population of owned (confined) dogs, whereas hunting and roaming dogs had no antibodies to the virus (91).

4.2. Endemic Viral Zoonoses

Whereas the big three diseases—HIV/AIDS, tuberculosis, and malaria—and some emerging viral zoonoses like AI and COVID-19 have dominated the global health agenda and attracted the most funding in recent years, endemic viral zoonoses have not received similar attention from donors, policy makers, or public health officials, with the consequence that their true impact on human and animal health is not known. Indeed, Maudlin et al. (92) noted that, unlike newly emerging zoonoses that attract the attention of the developed world, many endemic zoonoses are neglected, and this artificially reduces their importance in the eyes of administrators and funding agencies. This situation may be attributable to the fact that endemic zoonoses occur mainly in the developing world, where conditions for their maintenance and spread exist and may occasionally give rise to epidemics. In addition to affecting the health of people in the poorest communities, endemic zoonoses also negatively impact their livelihoods by lowering the productivity, or even causing the death, of their livestock. Unlike emerging zoonoses, the endemic zoonoses can be categorized as neglected diseases, and as a result, some are now reemerging health problems (40, 93). However, just like the emerging diseases, the endemic viral zoonoses, including rabies, CCHF, and RVF, also originate from animals.

4.2.1. Rabies. Rabies is an endemic, neglected, and grossly underreported zoonotic disease of poor communities and low- and middle-income countries that has become a reemerging global health threat (94, 95). It is a fatal viral disease that can infect all mammals, but domestic dogs constitute the predominant reservoirs, causing more than 99% of all human deaths from rabies

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(96), especially in parts of the world where domestic animal control and vaccination programs are limited (97). Safe and effective human vaccines are widely available for its prevention and control, and infection can be eliminated at the source through sustained mass vaccination of reservoir populations (98). However, rabies remains a neglected disease that is poorly controlled throughout much of the developing world, particularly Africa and Asia, where most human rabies deaths occur (97). Most industrialized nations have eradicated rabies from domestic dog populations, although it remains endemic and poorly controlled in domestic dog populations in most developing countries, with the majority (approximately 80%) of deaths occurring in rural areas (97, 99).

A disturbing dimension to the scourge of rabies is that with rising urbanization, the presence of traditional pets in households is increasing, and the licking of humans by domestic dogs that are rabid or suspected to be rabid poses a major risk to human health (87). In a study on the health and economic implications of dog rabies globally, Hampson et al. (94) reported that canine rabies causes approximately 59,000 human deaths, more than 3.7 million DALY, and US$8.6 billion economic losses annually, with the largest component of the economic burden due to premature death from rabies and costs of obtaining postexposure prophylaxis once an exposure has occurred. Rabies exemplifies a zoonotic disease that is transmitted by wild animals but has become adapted to urban environments. Bats, which colonize buildings; peri-domestic mammals like skunks and raccoons, which scavenge human refuse; and feral dogs commonly found on the streets in many countries are potential sources of rabies to domestic dogs, which could then become a major source of spillover infection to their human owners (100, 101).

4.2.2. Crimean-Congo hemorrhagic fever. CCHF virus (CCHFV) (genus *Orthonairovirus*, family *Nairoviridae*) is a tick-borne zoonotic virus that is endemic in Africa, Asia, and Southern and Eastern Europe (102). Ixodid ticks serve as reservoirs and vectors for CCHFV, and a variety of domestic animals, including cattle, sheep, goats, and camels, are its amplifying hosts. CCHF constitutes a significant public health threat, which can have huge impacts on abattoir workers and healthcare personnel, especially in resource-poor countries (103). In Nigeria, where there had been no reports on CCHFV circulation for more than three decades, Oluwayelu et al. (104) recently detected virus-specific immunoglobulin G (IgG) in 24% and 2% of tested cattle and goats, respectively, and concluded that these domestic species are a public health risk, especially to occupationally exposed individuals, such as veterinarians, butchers, and livestock farmers. In Ghana, Akuffo et al. (105) described exposure to CCHFV in slaughterhouse workers, and the virus was detected in ticks from cattle. The detection of CCHFV antibodies in domestic animals is reported to be useful because it provides evidence of circulating virus, identifies the location of CCHFV foci, and highlights a potential and increased risk for human infection (106). These findings underscore the need for continuous surveillance and targeted prevention and control strategies.

5. SURVEILLANCE FOR ZOONOTIC VIRAL DISEASES IN DOMESTIC LIVESTOCK POPULATIONS

Surveillance to monitor the occurrence of endemic, emerging, and reemerging viral zoonoses and their transmission is an important responsibility of public health authorities at local, national, regional, and global levels. Routine surveillance for zoonotic viral diseases in domestic livestock has become essential because of the critical role domestic animals play in the transmission of these diseases to humans, especially considering their increased interaction with humans, growing importance in human diets, and expanding populations worldwide. Currently, the world cattle, sheep, goat, pig, horse and chicken populations are estimated at 1.5 billion, 1.2 billion, 1 billion,

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1 billion, 58 million, and 24 billion, respectively (107). At various times in history, cattle, sheep, goats, poultry, pigs, horses, and camels have served as sources of introduction of viruses that pose great threats to public health. For instance, in 2003, an outbreak of high-pathogenicity AI (HPAI) subtype H7N7 in humans was linked to multiple commercial poultry farms in the Netherlands (108). Similarly, in the spring of 2009, a new influenza A (H1N1) virus emerged and spread rapidly throughout the world, with the WHO declaring it a pandemic on June 11. Further epidemiological investigation revealed that the virus that caused the human outbreak was related to swine influenza A (H1N1) viruses that circulated in pigs in North America and Europe/Asia and carried a mixture of genes from viruses circulating in those two geographical regions (109, 110). Active surveillance for these and other animal-origin influenza A viruses in livestock populations worldwide should be sustained to avert their emergence/reemergence in the human population.

Surveillance for AI at the human–animal interface is critical to identifying potential pandemic and zoonotic threats. One such interface is live-bird markets (LBMs), which present optimal conditions for the zoonotic transfer and evolution of infectious disease pathogens, as they are major contact points between humans and live (domestic and wild) animals and play a major role in facilitating emergence or reemergence of influenza viruses (111). This is buttressed by the findings of surveillance and molecular epidemiologic studies conducted at LBMs, which detected both low- and high-pathogenic AIVs in these markets, making them potential risk environments for transmission of infection to humans and other animals (112–114). Specifically, the risky practices of traders and butchers in the surveyed LBMs underscore the need for routine sero-epidemiologic and virologic surveillance of birds and occupationally exposed persons in these markets as an early-warning system (113). Further, other workers (115) reported that it is common to find free-range poultry from households scavenging for food close to or in the LBMs, thus making LBMs a hot spot for the introduction and spread of emerging AIV strains to commercial farms. Recently, low-pathogenicity AI (LPAI) H9N2 viruses recovered from apparently healthy poultry species during recent active surveillance conducted in LBMs across Nigeria were shown by whole-genome sequencing to belong to the G1 lineage, which has zoonotic potential (114).

6. DOMESTIC LIVESTOCK AS SENTINELS OF VIRAL ZOONOSES

Sentinel species are animals and plants that serve as harbingers of danger to human health and the environment. They provide insight into ecosystem function, identify hidden risks to human health, and predict future change (116, 117). Historical records show that canaries were used in the early twentieth century as indicators of the presence of odorless carbon monoxide in coal mines, as the bird would die first, giving miners sufficient time to escape (116). In the 1950s in Japan, cats were used as sentinels to detect mercury-contaminated fish, which caused deaths in humans consuming such fish (116). As with the analogy of the canary in the coal mine, domestic livestock reservoirs of some zoonotic viruses can constitute sentinels or early-warning systems that prompt the implementation of emergency response strategies aimed at either preventing the occurrence or curtailing the spread of such zoonoses in the human population. Early recognition and intervention during an emerging infectious zoonotic disease event are essential to limit spread (118). The death of American crows (*Corvus brachyrhynchos*), for instance, was reported as being critical in identifying West Nile virus (WNV) as the cause of the human outbreak of encephalitis and meningitis that occurred in late summer of 1999 in New York (119). Further, domestic Galliformes such as chickens and turkeys become viremic and develop high-titered antibody responses, allowing their use as sentinel animals (120). Apart from Unlu et al. (121), who reported that human WNV cases preceded sentinel chicken seroconversion in Louisiana, USA, other groups in Romania and Greece concluded that seroconversion in domestic chickens was

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widespread and that sentinel programs using domestic chickens would be useful for monitoring WNV activity (122, 123). Various factors such as successful selection of virus amplification sites with high enzootic transmission, testing of enough flocks and sentinels, and early detection of seroconversion by applying a sensitive diagnostic test may influence the capacity of sentinel chickens to function as early-warning indicators (123). Apart from domestic poultry, WNV antibodies have also been detected in dogs and horses, suggesting a role for these animals as sentinels for human risk due to WNV transmission (124, 125).

Detection of AIV antibodies in domestic animals that interact closely with humans, such as dogs (90, 126), is a good early-warning system for implementing preventive measures against outbreaks in humans, as these animals can serve as sentinels. Keatts et al. (127) opined that wild animals and domestic dogs can serve as sentinels for zoonotic diseases and other health concerns, and it appears essential that existing veterinary and wildlife surveillance systems for zoonotic pathogens are integrated with public health surveillance to better control such pathogens before they affect human health. We conclude that the possibility of humans acquiring influenza virus infection from direct contact with infected dogs and other domestic animals warrants concern and underscores the need for monitoring these animals during future AI outbreaks.

7. SPILLOVER EVENTS BETWEEN HUMANS AND DOMESTIC ANIMALS/WILDLIFE: IMPLICATIONS FOR OCCURRENCE OF VIRAL ZOONOSES

The spillover events that lead to disease emergence at the wildlife–livestock interface are driven by human-induced activities and exposure to domestic animals (128). Although the events (drivers) that result in the successful crossover of an animal microbe into humans are not well characterized, they are often precipitated by changes to ecological or biological systems, including altered patterns of contact between wild and domestic animals (e.g., NiV), direct human and wild animal contact (e.g., Ebola virus), and changes in species abundance or diversity (e.g., Hantavirus) (46). Also, Jones et al. (3) found a correlation between the emergence of zoonotic pathogens and the diversity of mammalian wildlife species. Indeed, animal species diversity, including the diversity of insect vectors and pathogenic microorganisms, is reported to increase toward the equator (129). This finding is supported by the report that regions where animal host species and pathogen richness are particularly high, and thus could trigger spillover to human hosts, exist within the 30° latitudinal and longitudinal bands (130).

Cross-species transmission of diseases between humans, domestic animals, and wildlife poses an increasing threat to human and animal health (44). The risk of spillover of zoonotic diseases from wildlife increases when wildlife–human and wildlife–domestic animal–human contacts increase (3, 43). For example, some authors observed higher biodiversity along ecosystem edges (e.g., along roads built into pristine ecosystems, or fragmented forest edges) where differing ecological systems meet, resulting in increased contact opportunities between a wider range of different host and pathogen species and increased potential pathways for spillover (131, 132).

Recent encroachments and anthropogenic pressures imposed on the natural habitats of wildlife have increased the interactions between people and animals and, consequently, human exposure to sylvatic disease cycles and the risk of pathogen spillover (3, 12). Additionally, expansion of livestock production, especially in proximity to wildlife habitats, has facilitated pathogen sharing in which livestock serve as amplifying hosts for zoonoses in humans (43, 44).

Several zoonotic viruses pathogenic for humans emerge through cross-species transmission from domestic or wild animals. According to Reusken et al. (133), there is substantial evidence that the MERS-CoV outbreak was fueled by zoonotic transmission from dromedary camels. This is largely because closely related viruses have been isolated from this but not any other animal

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species. Given the widespread geographical distribution of dromedaries found seropositive for MERS-CoV, continued transmission may occur in the future. The NiV outbreak in Malaysia and Singapore in 1998–1999 is a good example of the influence of intensification of livestock production on cross-species transmission of zoonoses. Once NiV crossed from wild bats to domestic swine, an explosive outbreak in high-density swine farms resulted in widespread exposure of humans and more than 250 human cases of encephalitis (134).

Another instance where livestock production intensification led to zoonosis emergence is the cross-species transmission of HPAI subtype H5N1 from wild aquatic birds (the natural reservoirs of influenza A viruses) to humans via massive amplification in domestic poultry (46). The interspecies transmission of influenza A virus is a crucial feature of its ecology and epidemiology. Marschall & Hartmann (135) noted that since HPAI virus subtype H5N1 was first detected in 1996 in domestic geese in China, it has crossed the species barrier and is now known to have infected humans, domestic cats, dogs, pigs, and other laboratory and wildlife species. These reports indicate an increasing need for surveillance of different animal species, especially those that enjoy a close relationship with humans, such as dogs, cats, domestic poultry, pigs, and horses, to better define their role in the epidemiology and ecology of influenza A viruses.

8. VIRAL ZOONOSES AND THE ONE HEALTH CONCEPT

To eliminate the potent threat that zoonoses constitute to human and animal health, there is need for institution of universally acceptable measures aimed at preventing pathogen spillover and reducing cross-species transmission at the animal–human–ecosystem interface and minimizing the health and economic impacts of these diseases. This necessitates a change in basic assumptions from the present siloed concept of addressing health challenges to a more integrated and holistic One Health (OH) approach, which recognizes the interconnectedness between the health of humans, animals, and their shared environment in the management and mitigation of the risks of EID. The OH approach is multi-sectoral and transdisciplinary, as it is based on interactions among physicians, veterinarians, entomologists, epidemiologists, microbiologists, agricultural scientists, environmentalists, anthropologists, sociologists, policy makers, and allied public health professionals, working collaboratively to improve and promote human and animal health (136, 137). These transdisciplinary collaborations are important for improving understanding of pathogen dynamics at the human–animal–environment interface and can provide opportunities for prevention or early detection and control of zoonotic disease threats before they become widespread outbreaks (138).

Organizations, consortia, and governments have adopted the OH approach for surveillance and control of several zoonoses. For instance, rabies control is a top priority for the WHO, the UN Food and Agriculture Organization, the World Organisation for Animal Health, and the Global Alliance for Rabies Control. These agencies have set the goal of eliminating dog-mediated human rabies by 2030 (139). In addition, the US Centers for Disease Control and Prevention developed a One Health Zoonotic Disease Prioritization (OHZDP) process that aims to overcome the challenges of zoonotic disease threats by identifying collaborative priorities and strengthening multi-sectoral collaboration at subnational, national, and regional levels (140). The OHZDP process includes development of policies and action plans to address the priority zoonotic diseases in collaboration with OH partners. For instance, using the OHZDP process to address zoonotic disease threats in the subregion, the Economic Community of West African States recently identified seven priority zoonoses for the region: anthrax, rabies, Ebola and other viral hemorrhagic fevers (e.g., Marburg fever, Lassa fever, RVF, CCHF), zoonotic influenzas, zoonotic tuberculosis, trypanosomiasis, and yellow fever. Adoption of the OH approach to address the subregion's priority zoonotic diseases in thematic areas including OH collaboration and coordination, surveillance and

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laboratory diagnosis, response and preparedness, prevention and control, workforce development, and research was recommended (141).

The success of this multidisciplinary approach has been driven by the synergistic impact of combining detailed and logistically challenging field sciences (e.g., ecology, field biology) with analytical approaches (e.g., epidemiological modeling, pathogen phylogenetic analysis) and laboratory science (e.g., serology, pathogen diagnostics, immunology) (142). Recently, to overcome human, animal, and environmental health challenges in Rwanda, Nyatanyi et al. (143) developed an innovative OH strategic plan that offers cooperation across systems, resulting in improved communication, evidence-based solutions, development of a new generation of systems thinkers, improved surveillance, decreased lag time in response, and improved health and economic savings.

9. METAGENOMICS FOR IDENTIFICATION OF ZOONOTIC VIRUSES IN POTENTIAL DOMESTIC ANIMAL RESERVOIRS

Viral metagenomics is the study of collective viral genomes from primary samples, e.g., environmental samples and clinical material from humans, animals, and insect tissue. This newly developed, culture- and sequence-independent method has been able to detect viruses behind diseases of unknown etiology, as well as allow the characterization of the complete viral populations in each sample (144). The metagenomics of viruses began in 2002 with the publication of two uncultured marine viral communities (145). However, recent advances in high-throughput, next-generation sequencing (NGS) technologies provide rapid and effective ways to identify the presence and diversity of novel viruses (146) without prior knowledge of the pathogens' genomic sequence data. NGS can also serve to increase the power of surveillance systems to detect infrequent zoonotic transmissions that could become pandemics (147).

The massive expansion and intensification of livestock farming worldwide have resulted in increased contact between humans and livestock, especially cattle, sheep, goats, pigs, and chickens, which can act as reservoirs for EID. Hence, detection and identification of potential zoonotic pathogens in these animals are important for public health surveillance and development of early-warning systems. In a study designed to characterize viral diversity in serum samples of domestic pigs, Masembe et al. (148) used pyrosequencing to show that these pigs were potential vertebrate hosts of Ndumu virus, a mosquito-borne alphavirus. They concluded that Ndumu virus represents a potential zoonotic agent, given the increasing risk of human–livestock–mosquito contact as the pig industry intensifies and the pig population increases in Uganda. Metagenomics has also been used to detect novel and known viruses with zoonotic potential in other domestic animals (149, 150).

Despite the many advantages of metagenomics in pathogen discovery, the detection of various viruses must be followed by other complementary investigations, such as attempting to recover the agent via virus isolation or transfection, and by performing pathogenicity and epizootiology studies to establish the Koch's postulates (151). Considering the wide range of detection of metagenomics techniques, i.e., their capacity to detect viruses either as single agents or as components in complex infections such as multifactorial infectious diseases, the discovery of a novel viral genomic sequence by viral metagenomics does not necessarily prove that the detected viruses are the causative agents of the studied diseases or disease complexes (151).

10. GAPS IN OUR KNOWLEDGE OF DOMESTIC ANIMALS AND VIRAL ZOONOSES

Current strategies for prevention and control of zoonotic diseases remain fragmented, with no coordination between the animal, human, and environmental health sectors. Hence, a strategic

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framework for prevention, control, and/or elimination of viral zoonoses at community, national, and regional levels should be developed with a view to minimizing the health, social, and economic impact of these diseases before they snowball into pandemics. This approach requires better understanding of the intricate interrelationships between domestic animal, human, and environmental health risks and outcomes that are crucial for development of early-detection systems and risk-reduction strategies to overcome the threats of endemic and novel viral zoonoses in the future. Further, the scarcity of surveillance data on several emerging viral zoonoses in many developing countries implies that the burden of livestock- and wildlife-origin zoonoses is underestimated and opportunities for control interventions are therefore limited.

Another knowledge gap is inadequate information on the diversity of zoonotic viruses transmissible by domestic animals, considering the close interaction between these animals and humans. This results in underestimation of the actual burden of disease they can spread. In addition, although zoonoses represent a significant emerging threat to public health, many of these diseases, such as diarrheal diseases arising from poor water sanitation, are neglected by funding agencies (152). The absence of cheap and effective vaccines against many zoonotic viral diseases, especially the emerging ones, is also a major barrier to attaining sustainable prevention and control of these diseases.

11. FUTURE PERSPECTIVES AND RECOMMENDATIONS

To better prepare for the prevention and control of zoonotic disease outbreaks, we must use the opportunities offered by the OH concept to identify, characterize, and understand viral zoonosis threats before they emerge. The data and information thus provided can be used to develop interventions and suitable countermeasures. Efforts aimed at promoting an OH approach for investigating these zoonotic viruses in their domestic animal reservoirs before they spill over to humans should be intensified. Moreover, the OH perspective on dealing with zoonotic disease threats from these animals would be more successful and cost effective if implemented with the full involvement of relevant stakeholders, especially medical professionals and policy makers. Expansion of regional, transdisciplinary OH networks that involve community members and diverse stakeholders, collation of real-time multifactorial data and explicit observations, and integration with public health networks can facilitate early-warning systems, rapid response, and mitigation for health threats to animals, humans, and the environment, while also advancing our understanding of endemic and emerging zoonotic health risks for affected communities (127).

Further, the development of integrated surveillance systems that link surveillance data of animals and humans and effectively model and predict human risk of zoonotic diseases should be prioritized. Miller et al. (153) noted that additional work is needed to establish the risk of exposure and transmission for pathogens of concern to humans and livestock and may necessitate surveillance studies elucidating potential risks for pathogens with the greatest transmission potential. In the context of emerging viral zoonoses, comprehensive risk assessments are needed to identify the animal–human and animal–animal interfaces where transmission of infectious agents occurs and the feasibility of risk-reduction interventions.

Another proactive approach to mitigating viral zoonosis threats is the use of epidemic intelligence, a tool that identifies key information on EID, specific incidents, and outbreaks and integrates both indicator- and event-based components in a single surveillance system (154, 155). It also permits continuous data gathering from the earliest stages through the evolution of an event, which informs dynamic risk assessments, guidance, and policy. Wilburn et al. (155) noted that epidemic intelligence is globally relevant, as it is central to the surveillance component of the International Health Regulations (156) to protect the global community from public

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health risks and emergencies that cross international borders. Developing a deeper understanding of the domestic livestock virome using metagenomics is crucial for detection of potential zoonotic and animal pathogens and emergency preparedness. A surveillance strategy based on next-generation technologies will be especially useful to contain diseases that may emerge within national boundaries, or that may potentially affect other countries. NGS can also increase the capacity of surveillance systems to detect infrequent zoonotic transmissions that could become pandemics (147).

Additionally, to mitigate the threats zoonotic viral diseases pose to human and animal health, and human livelihoods, global health organizations and national governments must improve funding for investigations designed to further elucidate the role of various domestic animals as potential reservoirs of high-consequence and priority zoonotic pathogens like AI and henipa- and coronaviruses. Such research, focused on determining the susceptibility of various domestic animals to infection with these viruses and animals' role in the epidemiology of the diseases, will be critical to informing appropriate animal and public health responses, including early warning–early response systems, to these disease threats. Given the potential for rapid zoonotic transmission of these viruses, routine surveillance activities are also needed to definitively determine the role of domestic animals in transmission of zoonotic viruses.

The possibility of vaccinating domestic animal reservoirs of zoonotic viruses using RNA-based vaccines, a new vaccine technology, can also be exploited. These vaccines use mRNAs, which, upon entering cells, are translated to antigenic molecules that in turn stimulate the immune system. RNA vaccines have been reported to elicit potent immune responses against cancers and infectious diseases, including SARS-CoV-2 (157, 158).

12. CONCLUDING REMARKS

The possibility of humans acquiring known and emerging zoonotic infections from direct contact with infected domestic animals warrants concern and underscores the need for regular monitoring of domestic animals, in and out of epidemic periods. Moreover, continuous surveillance efforts aimed at screening larger populations of exotic and domestic animals will assure a better state of preparedness for the prevention and control of zoonotic diseases.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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